

Add, subtract and multiply complex numbers

Basic skills

Q1. $z = 3 + 2i$, $w = 2 + i$, $v = 2 - 8i$

a) $z + w = 3 + 2i + 2 + i$ Add real,
Add imaginary
 $= (3 + 2) + (2 + 1)i$
 $= 5 + 3i$

b) $z - w = 3 + 2i - (2 + i)$
 $= (3 - 2) + (2 - 1)i$
 $= 1 + i$

c) $z \times w = (3 + 2i)(2 + i)$ double
brackets
 $= 6 + 4i + 3i + 2i^2$
 $= 6 + 4i + 3i - 2$ $i^2 = -1$
 $= 4 + 7i$

d) $w \times v = (2 + i)(2 - 8i)$

$$\begin{aligned}
 &= 4 - 16i + 2i - 8i^2 \\
 &= 4 + 8 - 14i \\
 &= 12 - 14i
 \end{aligned}$$

$$\begin{aligned}
 e) \quad w - v &= 2 + i - (2 - 8i) \\
 &= 0 + 9i = 9i
 \end{aligned}$$

$$f) \quad 5w = 5(2 + i) = 10 + 5i$$

$$\begin{aligned}
 g) \quad z - 2w &= 3 + 2i - 2(2 + i) \\
 &= 3 + 2i - 4 - 2i \\
 &= -1
 \end{aligned}$$

Competency skills

Q2. For $z = -3 + 5i$, $w = -5 - i$, $v = -2 + i$

$$\begin{aligned}
 a) \quad 2z + w &= 2(-3 + 5i) + -5 - i \\
 &= -6 + 10i - 5 - i = -11 + 9i
 \end{aligned}$$

$$\begin{aligned}
 b) \quad z - 2w &= -3 + 5i - 2(-5 - i) \\
 &= -3 + 5i + 10 + 2i = 7 + 7i
 \end{aligned}$$

$$\begin{aligned}
 c) \quad zw + v &= (-3+5i)(-5-i) + -2+i \\
 &= 15 + 3i - 25i - 5i^2 - 2 + i \\
 &= 13 - 21i + 5 = 18 - 21i \quad \text{(\textcolor{brown}{\downarrow} } i^2 = -1 \text{)}
 \end{aligned}$$

$$\begin{aligned}
 d) \quad w^2 + z &= (-5-i)^2 + -3+5i \\
 &= 25 + 10i + i^2 - 3 + 5i \\
 &= 22 + 15i - 1 = 21 + 15i
 \end{aligned}$$

$$\begin{aligned}
 e) \quad v^2 - w &= (-2+i)^2 - (-5-i) \\
 &= 4 - 4i + i^2 + 5 + i \\
 &= 9 - 3i - 1 = 8 - 3i
 \end{aligned}$$

$$\begin{aligned}
 f) \quad (2z)^2 &= 4z^2 = 4(-3+5i)^2 \\
 &= 4(9 - 30i + 25i^2) \\
 &= 4(-14 - 30i) \\
 &= -56 - 120i
 \end{aligned}$$

as in f)

$$\begin{aligned}
 g) \quad 3z^2 + 2z - 4 &= 3(-3+5i)^2 + 2(-3+5i) - 4 \\
 &= 3(-14 - 30i) - 6 + 10i - 4 \\
 &= -42 - 90i - 10 + 10i = -52 - 80i
 \end{aligned}$$



Q 3. $z = 3i - 1$

$$\begin{aligned} z^2 - 2z &= (3i - 1)^2 - 2(3i - 1) \\ &= 9i^2 - 6i + 1 - 6i + 2 \\ &= -9 - 12i + 3 = -6 - 12i \end{aligned}$$

Q 4. $z^3 = (5 + 2i)^3 = (5 + 2i)^2 (5 + 2i)$

$$= (25 + 20i - 4)(5 + 2i)$$

$$= (21 + 20i)(5 + 2i)$$

$$= 105 + 100i + 42i - 40$$

$$= 65 + 142i$$

real
part

so $\text{Re}(z^3) = 65$

imaginary
part

Q 5. $z^2 + 25 = (z + ai)(z + bi)$

$$z^2 + 25 = z^2 + (a + b)i z - ab$$

$$\text{so } a + b = 0 \quad -ab = 25$$

$$\text{so } a = 5 \quad b = -5$$

Alternatively $z^2 + 25 = 0$

$$z^2 = -25 \Rightarrow z = \pm 5i$$

so $(z - 5i)(z + 5i)$

Advanced skills

Q6. $(5+i)^2 = 25 + 10i - 1 = 24 + 10i$

$$\text{Im}((5+i)^2) = 10$$

$$\text{Im}(5+i) = 1$$

so no $\text{Im}(z^2) \neq \text{Im}(z)^2$

Q7. $z^3 - 3z + 25 = (z + ai)(z + bi)$
 $= z^2 + (a+b)i z - ab$

so $a+b = -3 \quad -ab = 25$

$$\text{so } a = -3 - b \Rightarrow (3+b)b = 25$$

$$3b + b^2 - 25 = 0$$

$$b^2 + 3b - 25 = 0$$

$$\text{So } b = \frac{-3 \pm \sqrt{109}}{2}$$

$$\text{So } a = -3 - \left(\frac{-3 \pm \sqrt{109}}{2} \right)$$

$$= \frac{-3 \mp \sqrt{109}}{2} \quad \text{notice } \mp \text{ is } \pm$$

$$\text{So } a = \frac{-3 + \sqrt{109}}{2}, b = \frac{-3 - \sqrt{109}}{2}$$

Q8. $z = 3 - 5i, w = 2i - 5$

a) $z^2 - 2iw = (3 - 5i)^2 - 2i(2i - 5)$

$$= 9 - 30i + 25i^2 - 4i^2 + 10i$$

$$= 9 - 30i - 25 + 4 + 10i$$

$$= -12 - 18i$$

b) $(i + w)^2 + z = (3i - 5)^2 + (3 - 5i)$

$$= -9 - 30i + 25 + 3 - 5i$$


$$= 19 - 35i$$

Q9.

$$zz^* = (a+bi)(a-bi)$$

$$= a^2 + abi - abi - bi^2$$

$$= a^2 + b^2$$

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